

Maxwell equations

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1 Definition of average energy

$$\overline{E} = \frac{\int_0^\infty n(E) E dE}{\int_0^\infty n(E) dE} \quad (1)$$

$$n(E) = n_0 \frac{2\pi}{(\pi kT)^{\frac{3}{2}}} \sqrt{E} e^{-\frac{E}{kT}} \quad (2)$$

2 Denominator first

$$\int_0^\infty n(E) dE \equiv F(E) = a n_0 \int_0^\infty \sqrt{E} e^{-bE} dE \quad (3)$$

$$a \equiv \frac{2\pi}{(\pi kT)^{\frac{3}{2}}} \quad (4)$$

$$b \equiv \frac{1}{kT} \quad (5)$$

$$\begin{aligned} x &= \sqrt{E} \\ 2x dx &= dE \end{aligned} \quad (6)$$

$$F(E) \rightarrow F(x) = a n_0 \int_0^\infty x e^{-bx^2} 2x dx \quad (7)$$

$$\int u dv = uv - \int v du \quad (8)$$

$$\begin{aligned} u &= x \\ du &= dx \end{aligned} \quad (9)$$

$$\begin{aligned} dv &= 2x e^{-bx^2} dx \\ v &= -\frac{1}{b} e^{-bx^2} \end{aligned} \quad (10)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x e^{-bx^2} + \frac{1}{b} \int_0^\infty e^{-bx^2} dx \right] \quad (11)$$

$$\begin{aligned} w &= x\sqrt{b} \\ \frac{1}{\sqrt{b}} dw &= dx \end{aligned} \quad (12)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x e^{-bx^2} + \frac{1}{b} \int_0^\infty e^{-w^2} \frac{1}{b} dw \right] \quad (13)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x e^{-bx^2} + \frac{1}{b^{\frac{3}{2}}} \int_0^\infty e^{-w^2} dw \right] \quad (14)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x e^{-bx^2} + \frac{1}{b^{\frac{3}{2}}} \frac{\sqrt{\pi}}{2} \int_0^\infty \frac{2}{\sqrt{\pi}} e^{-w^2} dw \right] \quad (15)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x e^{-bx^2} + \frac{1}{b^{\frac{3}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(w) \right] \quad (16)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x e^{-bx^2} + \frac{1}{b^{\frac{3}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(x\sqrt{b}) \right] \quad (17)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x e^{-bx^2} + \frac{1}{b^{\frac{3}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(x\sqrt{b}) \right] \Big|_0^\infty \quad (18)$$

$$F(E) = a n_0 \left[-\frac{1}{b} \sqrt{E} e^{-bE} + \frac{1}{b^{\frac{3}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(\sqrt{bE}) \right] \Big|_0^\infty \quad (19)$$

$$F(E) = a n_0 \left[\left(0 + \frac{1}{b^{\frac{3}{2}}} \frac{\sqrt{\pi}}{2} 1 \right) - (0 + 0) \right] \quad (20)$$

$$F(E) = \frac{2\pi}{\sqrt{\pi^3 k^3 T^3}} n_0 \left[\frac{k^3 T^3}{2} \frac{\sqrt{\pi}}{2} \right] \quad (21)$$

$$\int_0^\infty n(E) dE = n_0 \quad (22)$$

3 Now numerator

$$\int_0^\infty n(E) E dE \equiv F(E) = a n_0 \int_0^\infty \sqrt{E} E e^{-bE} dE \quad (23)$$

$$a \equiv \frac{2\pi}{(\pi k T)^{\frac{3}{2}}} \quad (24)$$

$$b \equiv \frac{1}{kT} \quad (25)$$

$$\begin{aligned} x &= \sqrt{E} \\ 2x dx &= dE \end{aligned} \quad (26)$$

$$F(E) \rightarrow F(x) = a n_0 \int_0^\infty x x^2 e^{-bx^2} 2x dx \quad (27)$$

$$F(E) \rightarrow F(x) = a n_0 \int_0^\infty x^3 e^{-bx^2} 2x dx \quad (28)$$

$$\int u dv = uv - \int v du \quad (29)$$

$$\begin{aligned} u &= x^3 \\ du &= 3x^2 dx \end{aligned} \quad (30)$$

$$\begin{aligned} dv &= 2x e^{-bx^2} dx \\ v &= -\frac{1}{b} e^{-bx^2} \end{aligned} \quad (31)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} + \frac{3}{b} \int_0^\infty x^2 e^{-bx^2} dx \right] \quad (32)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} + \frac{3}{b} \left(\int_0^\infty x x e^{-bx^2} dx \right) \right] \quad (33)$$

$$\begin{aligned} u &= x \\ du &= dx \end{aligned} \quad (34)$$

$$\begin{aligned} dv &= x e^{-bx^2} dx \\ v &= -\frac{1}{2b} e^{-bx^2} \end{aligned} \quad (35)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} + \frac{3}{b} \left(-\frac{1}{2b} x e^{-bx^2} + \frac{1}{2b} \int_0^\infty e^{-bx^2} dx \right) \right] \quad (36)$$

$$\begin{aligned} w &= x\sqrt{b} \\ \frac{1}{\sqrt{b}} dw &= dx \end{aligned} \quad (37)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} + \frac{3}{b} \left(-\frac{1}{2b} x e^{-bx^2} + \frac{1}{2b} \int_0^\infty e^{-w^2} \frac{1}{\sqrt{b}} dw \right) \right] \quad (38)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} + \frac{3}{b} \left(-\frac{1}{2b} x e^{-bx^2} + \frac{1}{2b^{\frac{3}{2}}} \int_0^\infty e^{-w^2} dw \right) \right] \quad (39)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} - \frac{3}{2b^2} x e^{-bx^2} + \frac{3}{2b^{\frac{5}{2}}} \int_0^\infty e^{-w^2} dw \right] \quad (40)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} - \frac{3}{2b^2} x e^{-bx^2} + \frac{3}{2b^{\frac{5}{2}}} \frac{\sqrt{\pi}}{2} \int_0^\infty \frac{2}{\sqrt{\pi}} e^{-w^2} dw \right] \quad (41)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} - \frac{3}{2b^2} x e^{-bx^2} + \frac{3}{2b^{\frac{5}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(w) \right] \quad (42)$$

$$F(x) = a n_0 \left[-\frac{1}{b} x^3 e^{-bx^2} - \frac{3}{2b^2} x e^{-bx^2} + \frac{3}{2b^{\frac{5}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(x\sqrt{b}) \right] \quad (43)$$

$$F(x) = a n_0 \left[-\frac{1}{b} \sqrt{E^3} e^{-bE} - \frac{3}{2b^2} \sqrt{E} e^{-bE} + \frac{3}{2b^{\frac{5}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(\sqrt{bE}) \right] \quad (44)$$

$$F(E) = a n_0 \left[-\frac{1}{b} \sqrt{E^3} e^{-bE} - \frac{3}{2b^2} \sqrt{E} e^{-bE} + \frac{3}{2b^{\frac{5}{2}}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(\sqrt{bE}) \right] \Big|_0^\infty \quad (45)$$

$$F(E) = a n_0 \left[(0 - 0 + \frac{3}{2b^{\frac{5}{2}}} \frac{\sqrt{\pi}}{2}) - (0 - 0 + 0) \right] \quad (46)$$

$$F(E) = a n_0 \frac{3}{2b^{\frac{5}{2}}} \frac{\sqrt{\pi}}{2} \quad (47)$$

$$F(E) = \frac{2\pi}{\sqrt{\pi^3 k^3 T^3}} n_0 \frac{3\sqrt{k^5 T^5}}{2} \frac{\sqrt{\pi}}{2} \quad (48)$$

$$F(E) = \frac{3\sqrt{k^5 T^5}}{\sqrt{k^3 T^3}} n_0 \frac{3}{2} \quad (49)$$

$$F(E) = \frac{3}{2} n_0 kT \quad (50)$$

$$\overline{E} = \frac{\int_0^\infty n(E) E dE}{\int_0^\infty n(E) dE} \quad (51)$$

$$\overline{E} = \frac{\frac{3}{2} n_0 kT}{n_0} \quad (52)$$

$$\overline{E} = \frac{3}{2} kT \quad (53)$$